A-A-2558
February 14, 1988

COMMERCIAL ITEM DESCRIPTION

DRILL, RATCHET (REVERSIBLE)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-D-698, Type I.

This commercial item description covers hand-operated, reversible ratchet drills with American standard self-holding taper sockets.

Salient characteristics.

Design. The ratchet drills shall consist essentially of a handle and head, body, ratchet mechanism, feed screw, pawl, and feed pin. Ratchet drills shall be furnished with a short spindle with feed screw which enables ratchet to be converted into a boiler ratchet or adapter, to use square taper shank drills. Ratchet drills, when tested as specified, shall be capable of withstanding the applicable torque loads. Handles and tapered sockets shall show no indication of permanent set, cracks, or flaws and the ratchet mechanism and ratchet reverse control pawl engaging surfaces shall show no damage.

Handle and head. The handle and head shall be either integral, threaded, or the head shall be welded securely to the handle. The handle shall be either a straight, round steel bar, or steel tubing smoothly finished to afford a comfortable handgrip. The head shall be drop-forged steel or malleable iron and all surfaces that bear or mate with the body shall have a satin finish.

Body. The body shall be drop-forged steel with a minimum of ten teeth or notches on the outside diameter to assure ratcheting with the handle having a swing of 36 degrees or less. The inside tapered hole shall be either American Standard taper or bitstock taper to accommodate drill shanks within the capacity of the drill. The tapered hole shall be smoothly finished and shall be concentric with the bearing surfaces on the outside diameter. The outside diameter of the spindle or the end opposite the tapered hole shall be threaded to accommodate a body nut, or other means provided for securely holding the body in the head. The body shall form a part of the ratchet mechanism. The taper square hole shall be smooth and concentric with the bearing surfaces on the outside diameter.

Sockets. Sockets for ratchet drills shall have a standard self-holding taper in accordance with ANSI B5.10.

Ratchet mechanism. The ratchet mechanism shall be enclosed in the head and secured in such a manner as to minimize loosening during extreme usage. The ratchet mechanism shall be reversible by movement of a shifting lever, knurled button, or equal means which permits ratcheting operation of the body in either direction of rotation. The shifting lever or button shall be steel and shall be located in the head or at the handgrip end of the handle.

Feed screw. The feed screw shall be alloy steel. The outside diameter of the screw shall be threaded with a class 3 thread for engaging the spindle end of the body. One end of the screw shall have a knurled collar with at least four holes equally spaced at right angles to the axis to provide means for applying pressure to the drill. The knurled end of the screw shall be provided with an integral or removable hardened-steel center.

Pawl. The pawl shall be hardened alloy steel. The pawl shall be firmly attached to a rod or pin suitable for reversing the action of the ratchet. A spring or other means shall be provided for applying the proper tension on the pawl when engaged in the teeth or notches of the body.

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Feed pin. The feed pin shall be steel, suitable for regulating the feed screw in order to apply pressure to the drill.

Screw threads. The screw threads used with the ratchet drills shall be in conformance with the screw-thread standards of FED-STD-H28.

Finish. All exposed metal surfaces of the ratchet drills shall have one of the coatings specified below. All external surfaces shall be free from pits, nodules, forge flash, cracks, burrs, and other detrimental defects.

Bright chromium plate. The coating shall be electro-deposited metals consisting of nickel followed by chromium. The minimum thickness shall be 0.0002 inch for nickel and 0.00001 inch for chromium.

Bright nickel plate. The coating shall be electro-deposited nickel. The minimum thickness shall be 0.0002 inch.

Zinc plate. The coating shall be electro-deposited zinc and shall be subjected to a chemical or electrochemical chromating. The minimum thickness shall be 0.0003 inch.

Hot phosphating. The coating shall consist of chemically produced phosphate, followed by a coating of preservative.

Oxide coating. The coating shall consist of chemically produced oxide, followed by a coating of preservative.

Sizes. See Table I.

Table I: American Standard self-holding taper socket.

National Stock Number	Size Number	Height of head (max) Inches	Length of feed (min) Inches	Length of handle (min) Inches	American Standard taper socket Number	Size of drill Inches	Torque load (min) Pound-Inch
5110-00-555-1784	1	5	2	9	1	1/16 to 9/16	2500
5110-00-223-5028	8	6-3/4	2-1/2	15	3	59/64 to 1-1/4	5500

Identification marking. Each drill shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number, and country of origin.

Workmanship. All drills shall be free from imperfections which may affect their safety, appearance, serviceability, or durability.

QUALITY ASSURANCE PROVISIONS

Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, Sampling Procedures and Tables for Inspection by Attributes, except where otherwise specified herein. The lot size shall consist of all drills offered for acceptance at one time. The sample unit is one drill.

Visual inspection. Each drill in the sample lot shall be examined for workmanship and marking requirements. The inspection level shall be III with an acceptable quality level (AQL) of 2.5 defects per hundred units.

Dimensional inspection. Each drill in the sample lot shall be examined for dimensional requirements. The inspection level shall be S-4 with an acceptable quality level (AQL) of 2.5 defects per hundred units.

Testing of end item. The inspection level shall be Level I with an AQL of 4.0 percent defective.

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Coating thickness test. Component parts of the ratchet drills shall be tested for coating thickness in accordance with the applicable test method in FED-STD-151, Section 500.

Torque load test. Ratchet drills shall be tested with the applicable torque load as specified in Table I. Handles shall be tested in conjunction with a tapered socket of the largest size as specified in Table I. The torque load shall be applied with a torque producing machine, lever with dead weights, or a testing machine. The load shall be applied for one minute and then removed. The effective lever arm, when the load is vertical and the lever is employed, is the shortest distance between two vertical lines passing respectively through the point of load application and the center of the taper shank fitting the body opening. The specific torque load shall be the product of the effective lever arm in inches and the applied load in pounds. After application of the torque load, the ratchet drill shall be inspected to determine compliance with the screw thread requirements.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, to the maximum extent practicable.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

The issue of referenced documents in effect on date of invitation for bid or request for proposal shall be used to determine compliance with the stated requirements. Sources for obtaining referenced documents may be determined by contacting the contracting officer.

ANSI standards are available from American National Standards Institute, 1430 Broadway, New York, NY 10018.

Federal standards are available from U.S. Government Printing Office, Washington, DC 20402.

Military standards are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

MILITARY INTERESTS:

Military Coordinating Activity
Army - GL

Custodians:
Army - GL
Air Force - 99

Review Activities:
Air Force - 84

PREPARING ACTIVITY:

GSA/FSS

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